

# A Clinical Benchmark of Public Self-Supervised Pathology Foundation Models

Corresponding Author: Dr Gabriele Campanella

**This file contains all reviewer reports in order by version, followed by all author rebuttals in order by version.**

Version 0:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

In their manuscript, the authors present a clinical benchmark for evaluating pathology foundation models. They use this dataset to benchmark four publicly available pathology foundation models and two additional models developed in-house. Notably, one of these models, Virchow, was published by the same research group in 2023.

I agree with the authors that an independent benchmark dataset is a valuable contribution for contextualizing newly published foundation models. While I understand that such a benchmark cannot be publicly released to prevent data leakage, the authors should explore methods for users to automatically benchmark their models on this dataset, such as through an API or a Docker-based system. Although the authors promise regular updates to the provided Git repository with evaluations of new foundation models, there is a risk that the repository could be abandoned if the researchers leave academia or shift focus to other projects.

The authors claim that multi-institutional pretraining data might improve the robustness and generalizability of foundation models. Similarly, I suggest that robustness and generalizability should also be evaluated using a diverse benchmark dataset. While the authors include a wide variety of tumor indications and clinical biomarkers, the institutional and technical variability (e.g., different scanners) is limited. This should be addressed in the discussion and potentially in future work.

In the introduction, the authors make several claims without providing supporting references, such as "Given the evidence from the natural language and vision domains that larger datasets and higher capacity models will produce better performance" and "Recent works show progress in this respect." Please add the corresponding references.

The authors observed that downstream performance in pathology does not scale with model and dataset size as it does in other domains. Can you think of reasons why the pathology domain behaves differently in this regard?

Minor comments:

- It took me some time to understand that in Table 4 "N" refers to the number of GPUs (and not batch size or samples). Please clarify this.
- Page 4: "there are still some challenges that need to overcome"
- Page 16: "aspects of pretraining a pathology foundation models"

(Remarks on code availability)

The code repository helps to understand how the feature embedding were used for linear probing using the GMA medal for slide-level aggregation. However, without dataset access the code cannot be used for the automatic evaluation of new foundation models. As stated above, I strongly encourage an API or a Docker-based system for automatic evaluation.

Reviewer #3

(Remarks to the Author)

1. Key features of the work

The authors collected a benchmark dataset to compare self-supervised pathology models. The dataset itself is the center

piece of this work.

## 2. Originality and significance:

While I agree on the necessity of a common benchmark dataset, the way the paper is written makes it difficult to identify the originality and significance of the manuscript itself beyond the dataset itself. My main concerns about the benchmark itself and its use in this study are:

- a. The dataset is not made available to public (understandably to prevent it being used as training, and I agree on that side) – The authors say “In the future, we plan to provide an API for users to benchmark their external models on our tasks.”. I think this would be crucial to make it available with this manuscript and would make this paper much more valuable.
- b. The dataset is not well documented: it doesn't seem to be fully described in detail as it is common in such studies (age, gender, survival curves when relevant, primary/met sites, stage etc...). Ideally, a full table with clinical data should be made available to have readers being aware of composition and potential biases. If those are used as benchmarks, there should be careful and complete descriptions of features compositions – if site-specific biases exist, users should be well aware of it (see “Howard, Frederick M., et al. “The impact of site-specific digital histology signatures on deep learning model accuracy and bias.” *Nature communications* 12.1 (2021): 4423.”).
- c. Also, related to this, apart from EGFR, because the dataset comes from a single institution, it also constitutes a very homogeneous source. It is unclear this particular dataset would indeed be a good representative and sufficient benchmark to compare multiple algorithms? I think “what would constitute a proper benchmark dataset” should be the first point discussed.
- d. Comparing foundation models is not easy. Because even though they learn features, they need some downstream classifier/task to achieve the desired goal. Here: “For this purpose we chose the popular Gated MIL Attention (GMA) model [20] with a linear classifier on top.”, but there is no discussion whether this is the best approach to compare those models. What's to say that another model would lead to different conclusions when comparing different SSL approaches? Furthermore, despite the MCCV split, the lack of another independent cohort makes it difficult to assess the classifier trained on top of the SSL.

## 3. Methodology:

In addition to adding more details on the dataset, the manuscript would benefit from clarifications on several points:

- a. “For each slide, tissue tiles were extracted at 20x magnification (0.5 microns per pixel, MPP) and embedded into a feature representation using a specific foundation model. Each slide is then converted to a 2D matrix where every row corresponds to a tile in the slide and the columns contain the features.”

Please clarify that the tile size and resolution match those used to train the different foundation models tested. As such, it is unclear if discrepancies during tiling or pre-processing would affect comparisons.

- b. “The exact parameters used for training can be found in the GitHub repository” - Since github pages may change over time, those used in the paper should be put in the method directly or in the supp data.
- c. “For Prov-GigaPath, the authors provide both a pretrained tile-level encoder and a pretrained slide-level aggregator, but since this work is focused on assessing the expressiveness of the feature representation of tile-level encoders, only the encoder portion of Prov-GigaPath is considered.”

Not sure this makes sense since your labels are per slide labels, not per tile. Also, it seems to contradict “The vectorized slide is the input to the GMA model which combines the tile representations into a slide-level representation which is then linearly projected to class scores.” which seems to say that a slide level classification is done.

- d. Figure 3: Ideally, this would only make sense if you were able to modify the number of training tiles while keeping all other parameters constant. So many other parameters affect those results (the architecture, the dataset composition for each category), that I am not sure how relevant this study is. How to be sure that other co-factors are not the source of those observations?

- e. Abstract and Table 1: The number of tiles will depend on the tile size and pixel resolution (magnification) used which should be specified here as well.

## 4. Conclusions/interpretation

I believe the following conclusions would need to be better supported or explained:

- a. “These results may be explained by the fact that lung is over-represented in the pretraining datasets of UNI and Prov-GigaPath. For UNI, lung is the second most common tissue in their dataset, with around 10% of the slides or about ten thousand WSIs.” Maybe a more complete summary (table?) of training vs test set would be interesting more generally for this section? For example, for all tests and each foundation model, check the overlap between the cohort (how much of the training set had slides from the tested cancer type and biomarker? Did the training set have any slides from the same scanner? Did the training set have any slide from the same institution?). Since types of tissues used, type of scanner and staining would be parameters affecting learning, these would be co-factors interesting to be aware of.
- b. “We have shown that the composition of the pre-training dataset may be a crucial aspect for explaining downstream performance.” As mentioned in previous comment, a more complete study should be done to really show it. I think you hint at it but should show it more thoroughly (as mentioned in the previous comment).
- c. “Among the pathology trained encoders, CTransPath consistently shows inferior performance.” This doesn't seem to be the case for bladder though. Any explanation why bladder would be different?
- d. “ImageNet pre-trained encoder is consistently under-performing the pathology trained encoders.” Please compute statistical significance to compare them.

## 5. References:

- a. “While there is unanimous agreement that DINO and DINOv2 perform better than other popular SSL algorithms for pathology pre-training, there is not much evidence to support a choice between the two.” Such statements should be argued with references. Were other tests and comparisons done? Where is this “unanimous agreement” from?

b. Although Kang et al paper's is mentioned (ref 25), since they also benchmarked SSL, maybe some more discussion / comparison with their work could be included.

c. "there are many academic and non-academic efforts to build a general-purpose pathology foundation model (Table 1)." I understand not all SSL network available can be tested depending on code and trained weights availability, but if this table aims to summarize all recently published foundations models, then it is incomplete (if it is done on a pre-defined selection, criteria for selection should be mentioned). Just a few examples of those which may have been missed: Lazard, T., Lerousseau, M., Decenciere, E. & Walter, T. Giga-ssl: Self-supervised learning for gigapixel images. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 4304–4313 (2023).

Li, Bin, Yin Li, and Kevin W. Eliceiri. "Dual-stream multiple instance learning network for whole slide image classification with self-supervised contrastive learning." Proceedings of the IEEE/CVF conference on computer vision and pattern recognition. 2021.

Claudio Quiros, Adalberto, et al. "Mapping the landscape of histomorphological cancer phenotypes using self-supervised learning on unannotated pathology slides." Nature Communications 15.1 (2024): 4596

Vu, Quoc Dang, et al. "Handcrafted Histological Transformer (H2T): Unsupervised representation of whole slide images." Medical image analysis 85 (2023): 102743.

d. Why cite arxiv papers when equivalent seems to be available in reviewed journal? (example for Vorontsov et al):

Vorontsov, Eugene, et al. "A foundation model for clinical-grade computational pathology and rare cancers detection." Nature Medicine (2024): 1-12.

#### 6. Clarity and context:

- a. "Tile-based predictions, organ classification, coarse segmentation, captioning, retrieval, and visual question answering (VQA) are valuable scientific explorations, but less relevant in the clinical setting." Maybe good to elaborate and clarify why you think so.  
VQA by pathologist seems highly relevant since they are the ones able to validate whether those foundation models make any sense from a pathological point of view.
- b. Figure 1 and 2: it would be best if y-axis of all plots were similarly scaled.

#### 7. Other comments / suggestions

- a. How does MSKCC LUAD EGFR perform on the trained network MSHS LUAD EGFR (and vice versa?)
- b. "Scaling Laws: Computational Resources" - what about inference? I guess users wouldn't retrain the network but run inference.

#### (Remarks on code availability)

I did not try to run the code, only looked at the github page which seems to be very minimalist and lacking information.

If I were to try to reproduce the results, I would need to clearly be able to see

- what scripts were used to tile the slides and with what options (not available on that github page)
- how data were organized for each SSL network (not available)
- what code the team run for the inferences for each network (there is some explanations on the "feature extraction" but not clear enough to be usable. What parameters are possible for the "encoder" option? What libraries need to be used?....)

Version 1:

Reviewer comments:

Reviewer #1

#### (Remarks to the Author)

I thank the authors for addressing the concerns raised in the initial review. I think by providing an automatic benchmarking pipeline, the submitted work significantly gained in importance for the community. For the manuscript, I still have some smaller comments that might require some alterations:

The introduction still includes many claims that are not backed by supporting references. E.g., "have enabled the training of predictive models for a variety of tasks with high performance" or "yielding results on par with supervised learning strategies" or "most studies in computational pathology rely on already existing pre-trained encoders". These are only some examples. I advise the authors to re-read their introduction and include supporting references where necessary.

Tables 2 and 3: Please include the full acronyms of the sites (MSHS, SUH, MSKCC) in the caption of all tables and not just

Tab. 1.

The PDF figure links are broken. When clicking on the link in "An overview of the workflow is presented in Supp. Figure 1.", the reader is re-directed to Figure 1 instead of supplementary figure 1. I also advise to include a section header for the supplementary material and rename the figures to Supp. Fig. 1 or Fig. S1.

I noticed that the authors significantly altered their author list by removing Thomas J. Fuchs as the second author. I assume this is related to my comment that the research group under Thomas J. Fuchs previously published the Virchow model which might introduce some conflict of interest. I think this change of authorship is not fully transparent and requires some explanation by the authors. Also, the benchmarking code is included in the repository of the "fuchs lab", which does indicate some connection to the research group.

Maybe, something similar to a "conflict of interest statement" should be added to the manuscript as the authors also mention in their manuscript that some of the benchmarked foundation models were pre-trained on data from the same clinical site and that it cannot be ensured that there is no overlap between the cohort and the clinical tasks based on MSKCC data.

(Remarks on code availability)

Reviewer #3

(Remarks to the Author)

Response to Q3e: I disagree with the authors saying that "the magnification information is not relevant in this context.". Since they do add graphs comparing the different methods, this is relevant. Especially when comparing and discussing less-well-explained prediction such as mutations and survival which may very well be resolution-dependent.

I thank the authors for adding section 5.6 on pre-training data composition.

Overall, while it is interesting and useful to compare the performances of different strategies, understanding why one is performing better can be tricky because of all the co-variables which can affect the interpretation (whether it is the size of the network, the composition of the training set, the magnification, or many other covariate which differentiate the various approaches). While I understand this is not the point of this study, the authors do dedicate a few section to try to explain those differences, and I wish the authors would nuance their conclusions and acknowledge more clearly of the difficulty to draw conclusion when potentially important co-variables are either not considered in the comparative analysis, or not known.

Response to Q6a:

It is unclear how this question was addressed. It is still unclear why the authors think so.

(Remarks on code availability)

Note: I did not to run the code but check it is available to some extent and instructions sound sufficient.

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# Rebuttal

We greatly appreciate the effort the reviewers put in analyzing our work. We have made several changes to address their concerns. Most notably, we now provide an automatic benchmarking pipeline that external users can take advantage of to benchmark their models on our clinical cohorts. We also expanded the set of foundation models and tasks included in our benchmark as well as extending the analysis to additional factors such as pretraining data composition and inference computational expenses. Finally, we made several modifications to improve rigor and clarity. We believe we have thoroughly addressed all the concerns from the reviewers, significantly strengthening the impact of this manuscript.

## Point by Point Response to Reviewers

### 1 Reviewer 1

Reviewer #1 (Remarks to the Author)

In their manuscript, the authors present a clinical benchmark for evaluating pathology foundation models. They use this dataset to benchmark four publicly available pathology foundation models and two additional models developed in-house. Notably, one of these models, Virchow, was published by the same research group in 2023.

**The authors of this work did not develop the Virchow model.**

I agree with the authors that an independent benchmark dataset is a valuable contribution for contextualizing newly published foundation models. While I understand that such a benchmark cannot be publicly released to prevent data leakage, the authors should explore methods for users to automatically benchmark their models on this dataset, such as through an API or a Docker-based system. Although the authors promise regular updates to the provided Git repository with evaluations of new foundation models, there is a risk that the repository could be abandoned if the researchers leave academia or shift focus to other projects.

**We absolutely agree with the reviewer that a mechanism to let users benchmark foundation models on their own would be valuable. Authors may even want to include results from our benchmark in their publications without the need of releasing their model publicly. We have already been working to accomplish this. We are now providing an automated benchmarking workflow which users can access by following the instructions provided in the GitHub repository linked in the manuscript. A description of the pipeline is provided in the methods section. The automatic benchmarking is run locally on the cluster, avoiding costs associated with cloud computing.**

The authors claim that multi-institutional pretraining data might improve the robustness and generalizability of foundation models. Similarly, I suggest that robustness and generalizability should also be evaluated using a diverse benchmark dataset. While the authors include a wide variety of tumor indications and clinical biomarkers, the institutional and technical variability (e.g., different scanners) is limited. This should be addressed in the discussion and potentially in future work.

We agree with the reviewer that our cohorts have relatively limited technical variability. This does not limit the relevance of the benchmark in providing insights on models' performance on clinically relevant tasks based on data generated during routine clinical workflows. Our work is the first benchmark in pathology to rely exclusively on clinically generated data. We addressed the limited technical variability in the limitations and future work sections of the manuscript.

In the introduction, the authors make several claims without providing supporting references, such as "Given the evidence from the natural language and vision domains that larger datasets and higher capacity models will produce better performance" and "Recent works show progress in this respect." Please add the corresponding references.

We thank the reviewer for noticing these faults. We corrected them by including appropriate supporting references in the manuscript.

The authors observed that downstream performance in pathology does not scale with model and dataset size as it does in other domains. Can you think of reasons why the pathology domain behaves differently in this regard?

This is an important question. We believe that one important reason is that just scaling the dataset size without performing any data selection may be detrimental to the end model. In most works so far the main drive is to include as much data as possible. This way the pretraining dataset may be highly enriched for homogeneous tissue with less variability.

Minor comments:

- It took me some time to understand that in Table 4 "N" refers to the number of GPUs (and not batch size or samples). Please clarify this. - Page 4: "there are still some challenges that need to overcome" - Page 16: "aspects of pretraining a pathology foundation models"

We addressed these mistakes in the manuscript.

Reviewer #1 (Remarks on code availability)

The code repository helps to understand how the feature embedding were used for linear probing using the GMA model for slide-level aggregation. However, without dataset access the code cannot be used for the automatic evaluation of new foundation models. As stated above, I strongly encourage an API or a Docker-based system for automatic evaluation.

We addressed this above.

## 2 Reviewer 3

1. Key features of the work The authors collected a benchmark dataset to compare self-supervised pathology models. The dataset itself is the center piece of this work.

2. Originality and significance: While I agree on the necessity of a common benchmark dataset, the way the paper is written makes it difficult to identify the originality and significance of the manuscript itself beyond the dataset itself. My main concerns about the benchmark itself and its use in this study are: a. The dataset is not made available to public (understandably to prevent it being used as training, and I agree on that side) – The authors say “In the future, we plan to provide an API for users to benchmark their external models on our tasks.”. I think this would be crucial to make it available with this manuscript and would make this paper much more valuable.

We agree with the reviewer that the availability of a mechanism for allowing external users to benchmark their own models would significantly strengthen this manuscript. To that end we have developed an automatic benchmarking pipeline which is described in the methods section of the manuscript. Instructions on how to use it are provided in the GitHub linked in the manuscript.

b. The dataset is not well documented: it doesn’t seem to be fully described in detail as it is common in such studies (age, gender, survival curves when relevant, primary/met sites, stage etc...). Ideally, a full table with clinical data should be made available to have readers being aware of composition and potential biases. If those are used as benchmarks, there should be careful and complete descriptions of features compositions – if site-specific biases exist, users should be well aware of it (see “Howard, Frederick M., et al. ”The impact of site-specific digital histology signatures on deep learning model accuracy and bias.” Nature communications 12.1 (2021): 4423.”).

We agree with the reviewer that including this information will enrich the manuscript. We have collected supporting information for each cohort and added relevant tables as part of the Supplementary material.

c. Also, related to this, apart from EGFR, because the dataset comes from a single institution, it also constitutes a very homogeneous source. It is unclear this particular dataset would indeed be a good representative and sufficient benchmark to compare multiple algorithms? I think “what would constitute a proper benchmark dataset” should be the first point discussed

The cohorts come from three institutions and while it can be argued that they present relatively limited technical variability, it is still a highly relevant dataset due to the fact that it is generated during routine clinical workflow. To the best of our knowledge this is the first benchmark to rely entirely on clinical data. We have included a comment in the limitations section of the manuscript regarding the limited technical variability. We have also included a statement regarding the significance of our benchmarking dataset in the introduction.

d. Comparing foundation models is not easy. Because even though they learn features, they need some downstream classifier/task to achieve the desired goal. Here: “For this purpose we chose the popular Gated MIL Attention



(GMA) model [20] with a linear classifier on top.”, but there is no discussion whether this is the best approach to compare those models. What’s to say that another model would lead to different conclusions when comparing different SSL approaches?

GMA is a good choice to assess pathology foundation models’ performance for several reasons. First, GMA is very simple, it does not use spatial information or tile-tile interaction. It relies entirely on the expressiveness of the tile-level representations which is exactly what we want to assess. Second, because of this, GMA is widely used to assess the performance of pathology foundation models. Third, while GMA is simple, it is highly performant. We have analyzed the performance of several recent aggregation strategies (Chen et al. 2024 - <https://arxiv.org/abs/2407.07841>), and in most cases, they do not outperform GMA. Finally, GMA is the most widely used aggregation strategy in computational pathology. We will expand on the selection of GMA, including relevant supporting literature.

Furthermore, despite the MCCV split, the lack of another independent cohort makes it difficult to assess the classifier trained on top of the SSL.

The MCCV strategy will correctly estimate the performance of the model if deployed in the institution of origin. Yet, for clinical implementation even in the institution of origin much more testing is required. In this manuscript we provide a benchmark to compare performance of foundation models and we consider out of scope to assessing the generalizability and clinical implementation of each model.

3. Methodology: In addition to adding more details on the dataset, the manuscript would benefit from clarifications on several points: a. ”For each slide, tissue tiles were extracted at 20x magnification (0.5 microns per pixel, MPP) and embedded into a feature representation using a specific foundation model. Each slide is then converted to a 2D matrix where every row corresponds to a tile in the slide and the columns contain the features.” Please clarify that the tile size and resolution match those used to train the different foundation models tested. As such, it is unclear if discrepancies during tiling or pre-processing would affect comparisons.

Yes, the resolution used in the downstream task experiments is appropriate for the foundation models used. We included a sentence for clarification.

b. ”The exact parameters used for training can be found in the GitHub repository” - Since github pages may change over time, those used in the paper should be put in the method directly or in the supp data.

We included the hyperparameters for training the in-house models in the Supplementary material.

c. ”For Prov-GigaPath, the authors provide both a pretrained tile-level encoder and a pretrained slide-level aggregator, but since this work is focused on assessing the expressiveness of the feature representation of tile-level encoders, only the encoder portion of Prov-GigaPath is considered.” Not sure this makes sense since your labels are per slide labels, no per tile. Also, it seems to contradict ”The vectorized slide is the input to the GMA model which combines the tile representations into a slide-level representation which is then linearly

projected to class scores.” which seems to say that a slide level classification is done.

All the tasks are slide level, but the encoders are tile level. In the GigaPath study the authors trained both a tile level encoder and a slide level aggregator. For the purposes of our benchmarks, only the tile level embedding model was used and used a GMA as aggregator as described. We rephrased the text to clarify this point.

d. Figure 3: Ideally, this would only make sense if you were able to modify the number of training tiles while keeping all other parameters constant. So many other parameters affect those results (the architecture, the dataset composition for each category), that I am not sure how relevant this study is. How to be sure that other co-factors are not the source of those observations?

We agree with the reviewer that many factors may influence downstream performance. Unfortunately, due to the expensive nature of training foundation models, a systematic comparison of all these factors has not been performed by any group and it is out of scope for this study. Instead we try to infer general trends based on the data available to us. While not an ablation study, our analysis highlights the current state of foundation models and hints at areas where improvement is needed.

e. Abstract and Table 1: The number of tiles will depend on the tile size and pixel resolution (magnification) used which should be specified here as well.

While it is true that the total number of tiles available from a set of slides depends on tile size and resolution, the actual number of tiles used for pretraining (reported in Table 1 from each respective study) is generally a sample of all available tiles. As such, we don’t think the magnification information is not relevant in this context.

4. Conclusions/interpretation I believe the following conclusions would need to be better supported or explained: a. ”These results may be explained by the fact that lung is over-represented in the pretraining datasets of UNI and Prov-GigaPath. For UNI, lung is the second most common tissue in their dataset, with around 10% of the slides or about ten thousand WSIs.” Maybe a more complete summary (table?) of training vs test set would be interesting more generally for this section? For example, for all tests and each foundation model, check the overlap between the cohort (how much of the training set had slides from the tested cancer type and biomarker? Did the training set had any slides from the same scanner? Did the training set had any slide from the same institution?). Since types of tissues used, type of scanner and staining would be parameters affecting learning, these would be co-factors interesting to be aware of.

We thank the reviewer for this comment. We have now included additional analyses which significantly strengthen our review of foundation models. In particular we have included an analysis of the prevalence of tissue within the pretraining dataset (whenever that information was available) and how it relates to downstream performance.

b. ”We have shown that the composition of the pre-training dataset may be a crucial aspect for explaining downstream performance.” As mentioned in

previous comment, a more complete study should be done to really show it. I think you hint at it but should show it more thoroughly (as mentioned in the previous comment).

Adressed in the previous comment.

c. "Among the pathology trained encoders, CTransPath consistently shows inferior performance." This doesn't seem to be the case for bladder though. Any explanation why bladder would be different? d. "ImageNet pre-trained encoder is consistently under-performing the pathology trained encoders." Please compute statistical significance to compare them.

For each task we performed pairwise statistical tests across all models in support of our result descriptions. Details are included in the text and test results are shared in the Supplementary material.

5. References: a. "While there is unanimous agreement that DINO and DINOv2 perform better than other popular SSL algorithms for pathology pre-training, there is not much evidence to support a choice between the two." Such statements should be argued with references. Were other tests and comparisons done? Where is this "unanimous agreement" from?

We modified this statement to be more objective and included relevant references.

b. Although Kang et al paper's is mentioned (ref 25), since they also benchmarked SSL, maybe some more discussion / comparison with their work could be included.

We expanded the comparison with related work in the introduction and discussion sections

c. "there are many academic and non-academic efforts to build a general-purpose pathology foundation model (Table 1)." I understand not all SSL network available can be tested depending on code and trained weights availability, but if this table aims to summarize all recently published foundations models, then it is incomplete (if it is done on a pre-defined selection, criteria for selection should be mentioned). Just a few examples of those which may have been missed: Lazard, T., Lerousseau, M., Decenciere, E. & Walter, T. Giga-ssl: Self-supervised learning for gigapixel images. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 4304–4313 (2023).

Li, Bin, Yin Li, and Kevin W. Eliceiri. "Dual-stream multiple instance learning network for whole slide image classification with self-supervised contrastive learning." Proceedings of the IEEE/CVF conference on computer vision and pattern recognition. 2021.

Claudio Quiros, Adalberto, et al. "Mapping the landscape of histomorphological cancer phenotypes using self-supervised learning on unannotated pathology slides." Nature Communications 15.1 (2024): 4596

Vu, Quoc Dang, et al. "Handcrafted Histological Transformer (H2T): Un-supervised representation of whole slide images." Medical image analysis 85 (2023): 102743.

We expanded the list of foundation model benchmarked to include some newer models as well as older but popular models. We expanded the description

of the inclusion criteria to emphasize that tile-level public encoders are the focus of this benchmark.

d. Why cite arxiv papers when equivalent seems to be available in reviewed journal? (example for Vorontsov et al):

Vorontsov, Eugene, et al. "A foundation model for clinical-grade computational pathology and rare cancers detection." *Nature Medicine* (2024): 1-12.

The journal publication was not available at the time of submission. We updated the reference with the journal article.

6. Clarity and context: a. "Tile-based predictions, organ classification, coarse segmentation, captioning, retrieval, and visual question answering (VQA) are valuable scientific explorations, but less relevant in the clinical setting." Maybe good to elaborate and clarify why you think so. VQA by pathologist seems highly relevant since they are the ones able to validate whether those foundation models make any sense from a pathological point of view.

With the renewed interest in vision-language models, we have modified the text to circumscribe the scope of this work to vision only models.

b. Figure 1 and 2: it would be best if y-axis of all plots were similarly scaled.

We prefer the current version which allows to better perceive the differences in performance between models. But we are open to modify the figure if necessary.

7. Other comments / suggestions a. How does MSKCC LUAD EGFR perform on the trained network MSHS LUAD EGFR (and vice versa?)

While interesting, we believe it is out of scope of this work to assess the aggregator generalization performance.

b. "Scaling Laws: Computational Resources" - what about inference? I guess users wouldn't retrain the network but run inference.

Inference is a topic that we did not consider in our original submission. Foundation model inference depends mostly on the encoder architecture, and since many of them use the same architectures, the inference time will be the same. Based on the reviewer's comment, we are including an inference section in the results.

Reviewer #3 (Remarks on code availability)

I did not try to run the code, only looked at the github page which seems to be very minimalist and lacking information. If I were to try to reproduce the results, I would need to clearly be able to see - what scripts were used to tile the slides and with what options (not available on that github page)

The code was meant to illustrate the benchmarking strategy, abstracting away from the data used since it is not made available. We have now described the slide processing steps in more detail, including the tiling algorithm.

- how data were organized for each SSL network (not available)

We included a brief description of how we structured the data.

- what code the team run for the inferences for each network (there is some explanations on the "feature extraction" but not clear enough to be usable. What parameters are possible for the "encoder" option? What libraries need to be used?....)

We clarified the options. These correspond to the datasets and foundation models included in the study.

# Response to Reviewers

## A Clinical Benchmark of Public Self-Supervised Pathology Foundation Models

Campanella et al.

We extend our sincere thanks to the Editor and Reviewers for their thoughtful and comprehensive evaluations of our manuscript, *A Clinical Benchmark of Public Self-Supervised Pathology Foundation Models*. Your insightful feedback has been invaluable in improving the clarity and rigor of our work. We appreciate the time and effort you have dedicated to this process, and we have carefully addressed each of your comments in this point by point response and in our revised manuscript. Thank you for your constructive critiques and for helping to strengthen the impact of our study.

### Reviewer 1

I thank the authors for addressing the concerns raised in the initial review. I think by providing an automatic benchmarking pipeline, the submitted work significantly gained in importance for the community. For the manuscript, I still have some smaller comments that might require some alterations:

**We thank Reviewer #1 for their thoughtful review and their help in significantly improving our study.**

The introduction still includes many claims that are not backed by supporting references. E.g., “have enabled the training of predictive models for a variety of tasks with high performance” or “yielding results on par with supervised learning strategies” or “most studies in computational pathology rely on already existing pre-trained encoders”. These are only some examples. I advise the authors to re-read their introduction and include supporting references where necessary.

**We are sorry for not having addressed this issue to a satisfactory level in our previous response. We have carefully gone through the introduction and we have rephrased or included citations where necessary. We believe we have now addressed this source of concern satisfactorily.**

Tables 2 and 3: Please include the full acronyms of the sites (MSHS, SUH, MSKCC) in the caption of all tables and not just Tab. 1.

**We have included the full acronyms.**

The PDF figure links are broken. When clicking on the link in “An overview of the workflow is presented in Supp. Figure 1.”, the reader is re-directed to Figure 1 instead of supplementary figure 1. I also advise to include a section

header for the supplementary material and rename the figures to Supp. Fig. 1 or Fig. S1.

We thank the reviewer for noticing these mistakes on our part. For final submission we have removed hyperlinks and have adhered strictly to the editorial policies regarding figure and table naming conventions.

I noticed that the authors significantly altered their author list by removing Thomas J. Fuchs as the second author. I assume this is related to my comment that the research group under Thomas J. Fuchs previously published the Virchow model which might introduce some conflict of interest. I think this change of authorship is not fully transparent and requires some explanation by the authors. Also, the benchmarking code is included in the repository of the "fuchs lab", which does indicate some connection to the research group.

We appreciate the reviewer's concerns regarding this subject and we want to make sure we are leaving no room for further concern. Dr. Fuchs served as the principal investigator of the computational pathology lab at Mount Sinai during the period when the bulk of the experiments for this study were conducted. It is important to clarify that Dr. Fuchs's academic work at Mount Sinai was entirely independent of his roles at Paige, a company he founded which has released some of the models we analyzed in our study. During the peer review process, Reviewer #1 raised a concern regarding the potential conflict of interest, given Dr. Fuchs's involvement with Paige. The reviewer mistakenly perceived that our team might have been involved with the training of Paige's foundation models due to Dr. Fuchs's association with the company. We wish to state unequivocally that Dr. Fuchs was not involved in the training of any foundation models at Paige. Dr. Vanderbilt holds a small equity stake in Paige, but has no operational role in the company, as disclosed in the Competing Interests section. No other team member has any relationship with Paige in any capacity. In light of Reviewer #1's comments and to prevent any potential misattribution of our team's efforts as being associated with Paige, Dr. Fuchs made the decision to voluntarily remove himself from the publication. This action was taken to uphold the integrity and clarity of our research and to eliminate any possible confusion regarding the source of our study's findings.

In addition, we have shared the code in a new repository managed by Dr. Campanella to avoid confusion and we have included a note in the discussion about the potential for the Virchow models to have used data during training that is included in this study.

Maybe, something similar to a "conflict of interest statement" should be added to the manuscript as the authors also mention in their manuscript that some of the benchmarked foundation models were pre-trained on data from the same clinical site and that it cannot be ensured that there is no overlap between the cohort and the clinical tasks based on MSKCC data.

We would be happy to include a statement if the editorial team deems this necessary.

## Reviewer 3

Response to Q3e: I disagree with the authors saying that “the magnification information is not relevant in this context.”. Since they do add graphs comparing the different methods, this is relevant. Especially when comparing and discussing less-well-explained prediction such as mutations and survival which may very well be resolution-dependent.

We understand the reviewer’s concerns. We stressed that all downstream tasks were performed with tiles extracted at 20x which corresponds to the magnification used for pretraining each of the models in this analysis. To drive this point more clearly, we have included the pretraining magnification in Table 1.

I thank the authors for adding section 5.6 on pre-training data composition.

We thank the reviewer for the idea of performing this analysis, strengthening the impact of the study.

Overall, while it is interesting and useful to compare the performances of different strategies, understanding why one is performing better can be tricky because of all the co-variables which can affect the interpretation (whether it is the size of the network, the composition of the training set, the magnification, or many other covariate which differentiate the various approaches). While I understand this is not the point of this study, the authors do dedicate a few section to try to explain those differences, and I wish the authors would nuance their conclusions and acknowledge more clearly of the difficulty to draw conclusion when potentially important co-variables are either not considered in the comparative analysis, or not known.

We fully agree with the reviewer’s perspective on the difficulty of deriving conclusions from the evidence at our disposal. We have modified the Discussion section to acknowledge these uncertainties.

Response to Q6a: It is unclear how this question was addressed. It is still unclear why the authors think so.

We are sorry for not properly addressing the reviewer’s point in our previous response. Question 6a was: “Tile-based predictions, organ classification, coarse segmentation, captioning, retrieval, and visual question answering (VQA) are valuable scientific explorations, but less relevant in the clinical setting.” Maybe good to elaborate and clarify why you think so. VQA by pathologist seems highly relevant since they are the ones able to validate whether those foundation models make any sense from a pathological point of view.

Tile-level predictions are less relevant clinically because a diagnosis is emitted at the slide level and a model’s performance on the tile level is not directly related to the performance in the diagnostic setting. In regards to the vision-language tasks such as captioning and VQA, quantifying model performance is difficult and still an open research question. As such, we leave the benchmarking of vision-language models to future work. But to answer the question, vision-language models have not yet reached the stage where clinical integration has been proposed or validated. This may be in part also due to the difficulty of quantifying performance. To keep the introduction concise, we have decided to remove this statement.



Reviewer #3 (Remarks on code availability):

Note: I did not to run the code but check it is available to some extent and instructions sound sufficient.